Uday Kalyan S

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Education

Indian Institute of Technology, Kharagpur Aug. 2024 – Present
B.Tech. in Artificial Intelligence *(CGPA: **8.89/10**)* *Kharagpur, WB*

Gopalan PU College (Karnataka State Board) June 2022 – May 2024
*Pre-University Course (Aggregate: **89.3%**)* *Bangalore, KA*

Publications

Kill the Rollouts: Online Caching Optimization for Shielding-Based Diffusion Planners **2026**
Uday Kalyan S, V. Buereddy, R. Prakash | Accepted in IEEE IROS 2026 Workshop: AIM-Ctrl

Internships

Research Intern - Human Interactive Robotics Lab, IISc Bangalore May 2026 – Jul. 2026
Advisor: Prof. Ravi Prakash | Safe Motion Planning, Diffusion Models

- Designed a **safety-guided diffusion planner** fusing lazy safety verification with model-based trajectory search.
- Improved on state-of-the-art baselines by **38%** via reachability-aware, feasibility-constrained trajectory generation.
- Built a parallelized **Python/JAX** simulation pipeline, benchmarking planning time on linear point-mass dynamics.

Technical Projects

Action Dependency Graph Optimization - open source contribution Dec. 2025 - June 2026
MARMot Lab, National University of Singapore | Multi-Agent Motion Planning

- Relaxed the **equal-robot-size assumption** of a published multi-agent planner via **AABB-tree** collision checks.
- Cut planner runtime by **77%** (0.88s to 0.199s) and coordination graph complexity from $O(N^4)$ to $O(N^3)$ at scale.
- Upstreamed the work into the **Open-RMF / MAPF** stack (80+ stars), removing heterogeneous-robot collisions.

Autonomous warehouse scanning robot - Inter IIT Tech Meet 14.0 Oct. 2025 - Dec. 2025
Contingent Gold | SLAM, Controls, Computer Vision, Real-time Robotics

- Engineered an **autonomous robot** to scan 2m racks via z-axis mechanisms, PID control and **lidar based SLAM**.
- Implemented a **distributionally robust CLF-CBF** for probabilistic avoidance of dynamic obstacles at **50 Hz**.

Invertible Neural Network (INN) for Localization Sept. 2025
Deep Learning, Robotics Localization

- Resolved the 'kidnapped robot' problem via **bijective pose mapping**, improving inference speed by 20% in **CARLA**.
- Built a custom **PyTorch** invertible network with a **VAE** for LIDAR compression and positional encoding of poses.

Wasserstein GAN for Image Generation - course project Jan. 2026 - Mar. 2026
Advisor: Prof. Kaustav Rudra | Deep Learning, Generative Modelling

- Built a **Wasserstein GAN** pipeline on **LSUN Bedroom**, using the Wasserstein distance for stable optimization.
- Compared **WGAN**, GAN, DCGAN and MLP architectures via convergence analysis and qualitative evaluation.

Positions of Responsibility

Planning & Controls Team Head - Autonomous Ground Vehicle Research Group Aug. 2026 – Present
Advisor: Prof. Debashish Chakravarthy | Research & Development in Robotics

- Configured **ROS2 SLAM Toolbox** to fuse Lidar data and build occupancy maps for the **AutoDrive** simulator.
- Integrated a **monocular visual odometry** pipeline in OpenCV for pose estimation and sparse 3D reconstruction.

Executive Head - Kharagpur Open Source Society Apr. 2026 – Present
Open Source Development & Community

- Organized **Kharagpur Winter of Code** for 300+ participants, managing contributor onboarding and the website.
- Conducted campus workshops on Linux, Git/GitHub and deployment workflows, promoting engineering practices.

Technical Skills

Languages & Tools: Python, C++, Rust, C, ROS2, SLAM Toolbox, Git, GitHub Actions, Docker, Linux, LaTeX
Libraries & Frameworks: PyTorch, JAX, NumPy, SciPy, Pandas, Scikit-learn, OpenCV, Matplotlib, Gym, CARLA

Relevant Coursework & MOOCs

AI & ML: Optimization for AIML, Deep Learning Theory, Linear Algebra for AIML, Introduction to ML, AI, Deep RL
Math & Systems: Advanced Calculus, Probability & Statistics, Algorithm Design & Analysis, Systems Programming